

# **Functional Imaging of Pelvic Floor Muscles**

**Ultrasound Scanning of Asymptomatic Subjects**

# Methodology

1. Purpose: To gain understanding of the mechanisms of pelvic floor action in terms of displacement velocity and acceleration
2. Perineal Ultrasound Scanning (5.5 Mhz Elliptical Probe)
3. Number of subjects: 23
4. Age Range:  $31.1 \pm 13.6$  years
5. Body Mass:  $22.4 \pm 1.9$  kg
6. Segment: Urethra Displacement, and Anorectal Junction Motility

# Data Collection Analysis

- Video Recordings of Imaging Signals being recorded on a PC for offline analysis by a trained physician
- Analysis of the data that was performed offline
- Velocity was analyzed by differentiating displacement/time
- Acceleration was computed by re-differentiating velocity/time

# Effect on bladder/urethra and effect on anorectal junction(ARJ)

## PFN(Active), Valsalva(Passive), Cough(Transient)

Voluntary Activation (PFM Precontraction / “The Knack”)

- **Bladder/Urethra:** Increases urethral closure pressure; elevates bladder neck; prevents urine leakage during stress
- **Anorectal Junction (ARJ):** Cranial lift and stabilization of pelvic floor structures

Passive Activation (Valsalva)

- **Bladder/Urethra:** Downward descent of bladder neck and urethra; increased risk of leakage/funneling
- **Anorectal Junction (ARJ):** Caudal (downward) displacement and organ descent

Transient Activation (Cough)

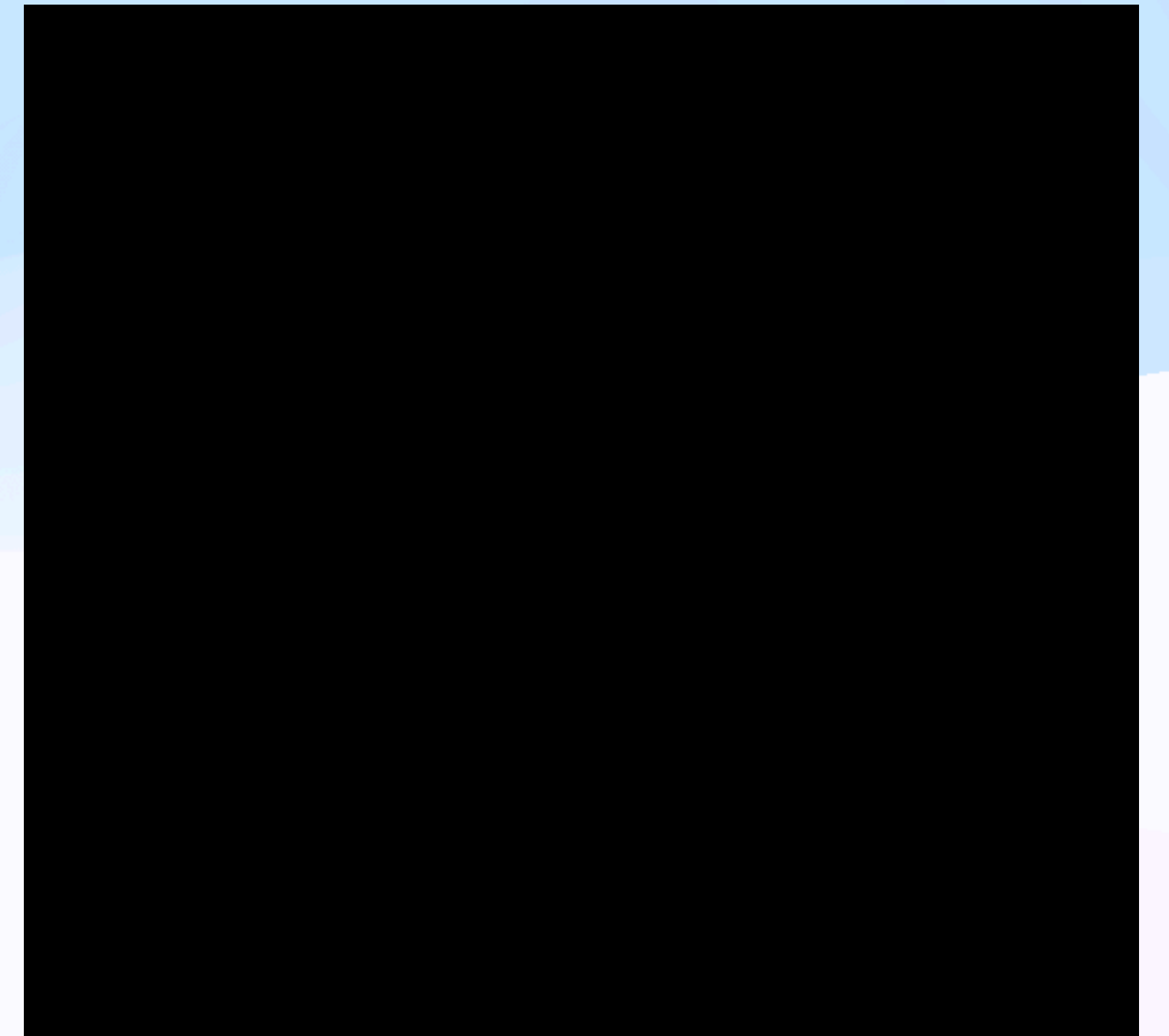
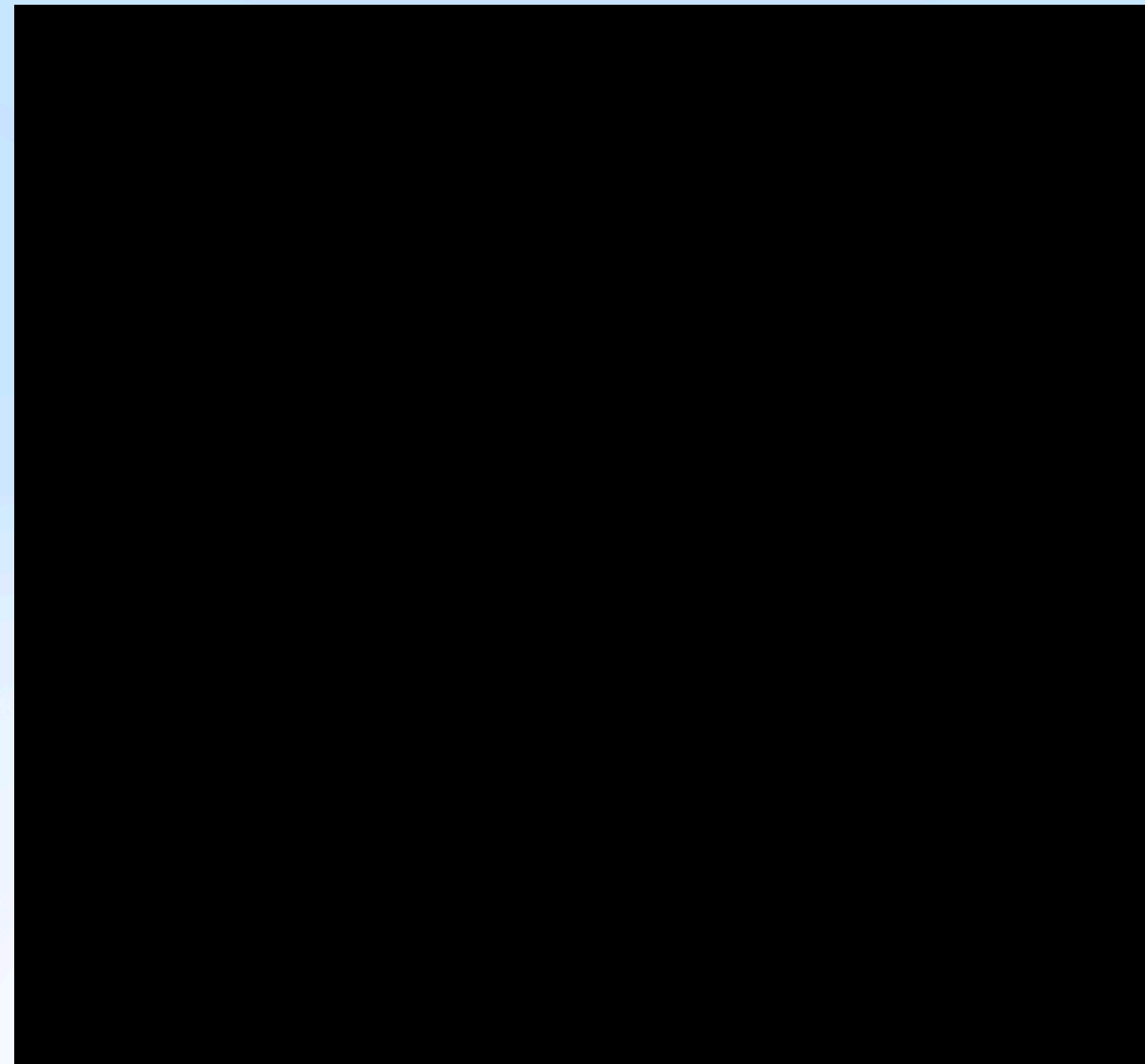
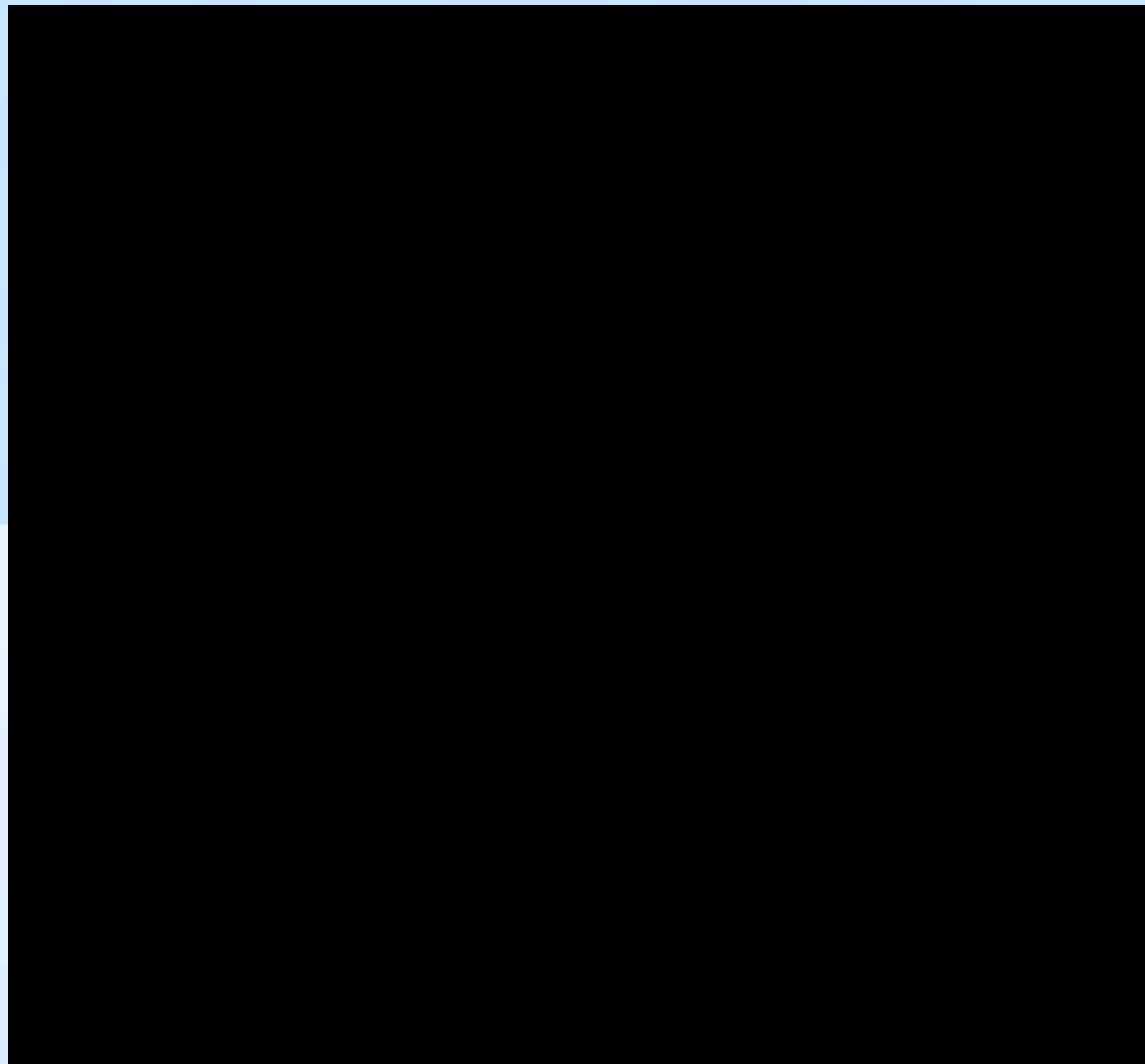
- **Bladder/Urethra:** Rapid rise in intra-abdominal pressure → urine loss without protection; “The Knack” reduces leakage by 73–98%
- **Anorectal Junction (ARJ):** Brief reflexive pelvic floor response; enhanced by pre-contraction

Key Finding (Miller et al., 1998)

- Timed PFM contraction (“The Knack”) before/during cough significantly reduces cough-related urine loss in women with mild SUI within 1 week

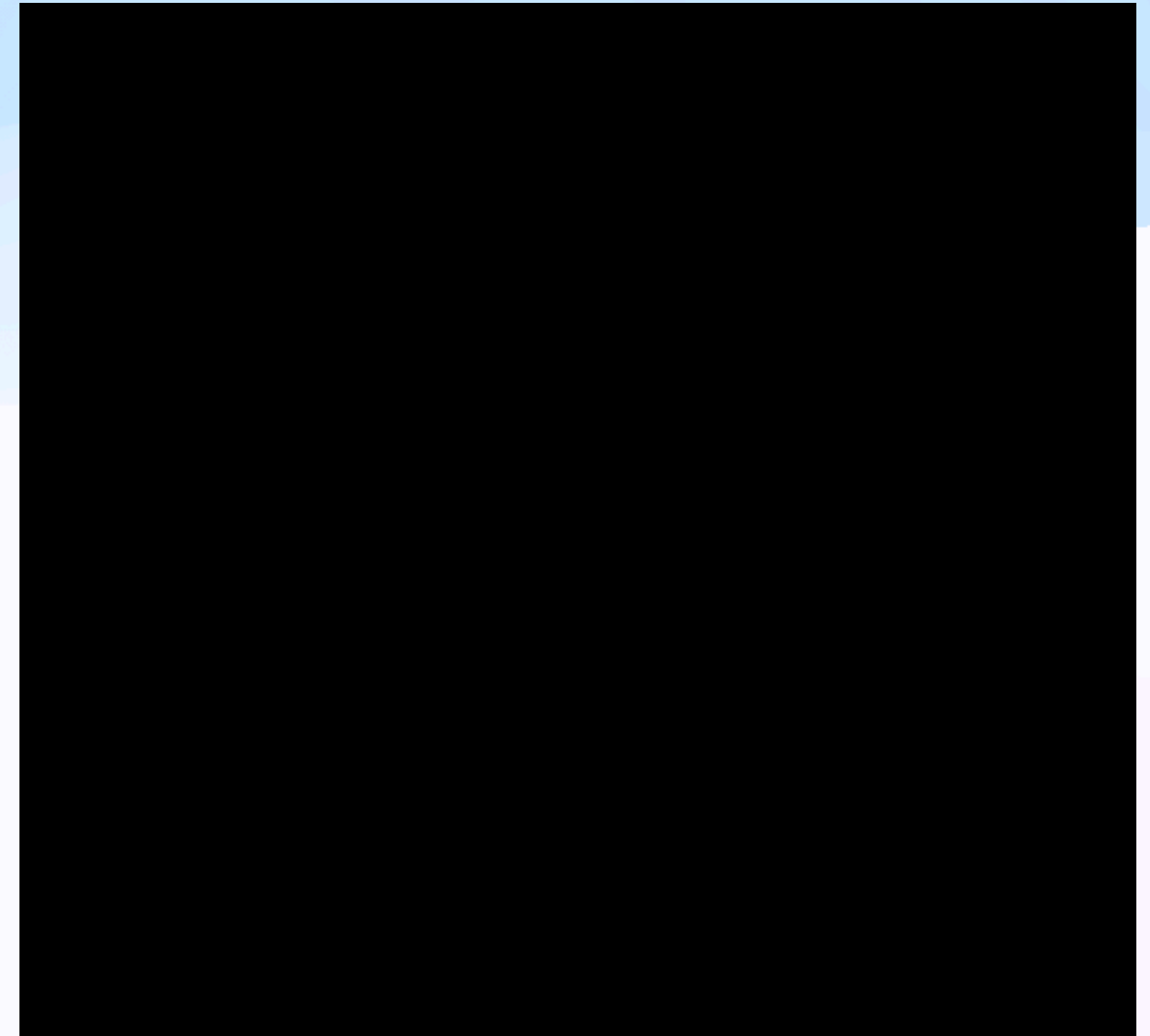
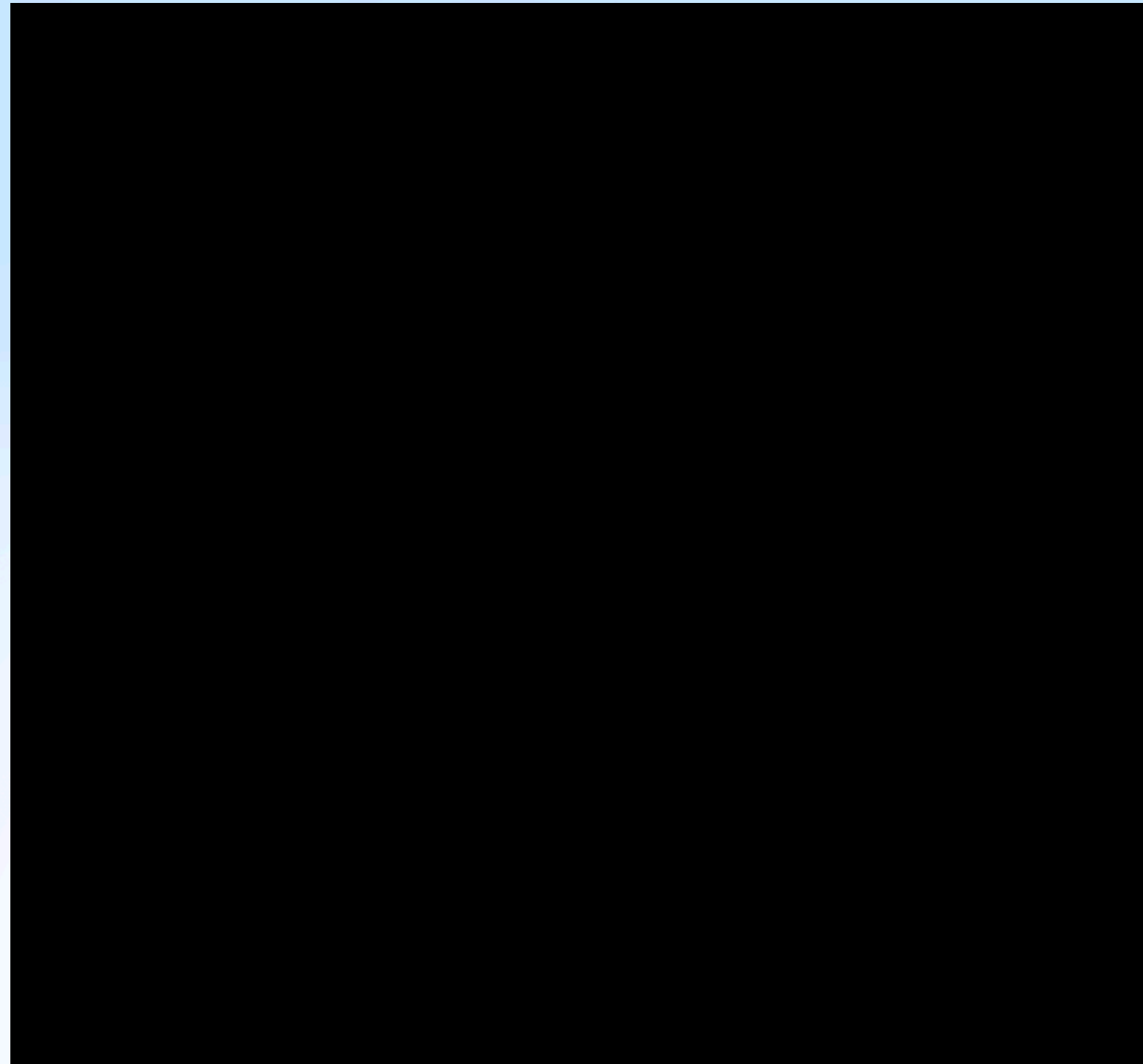
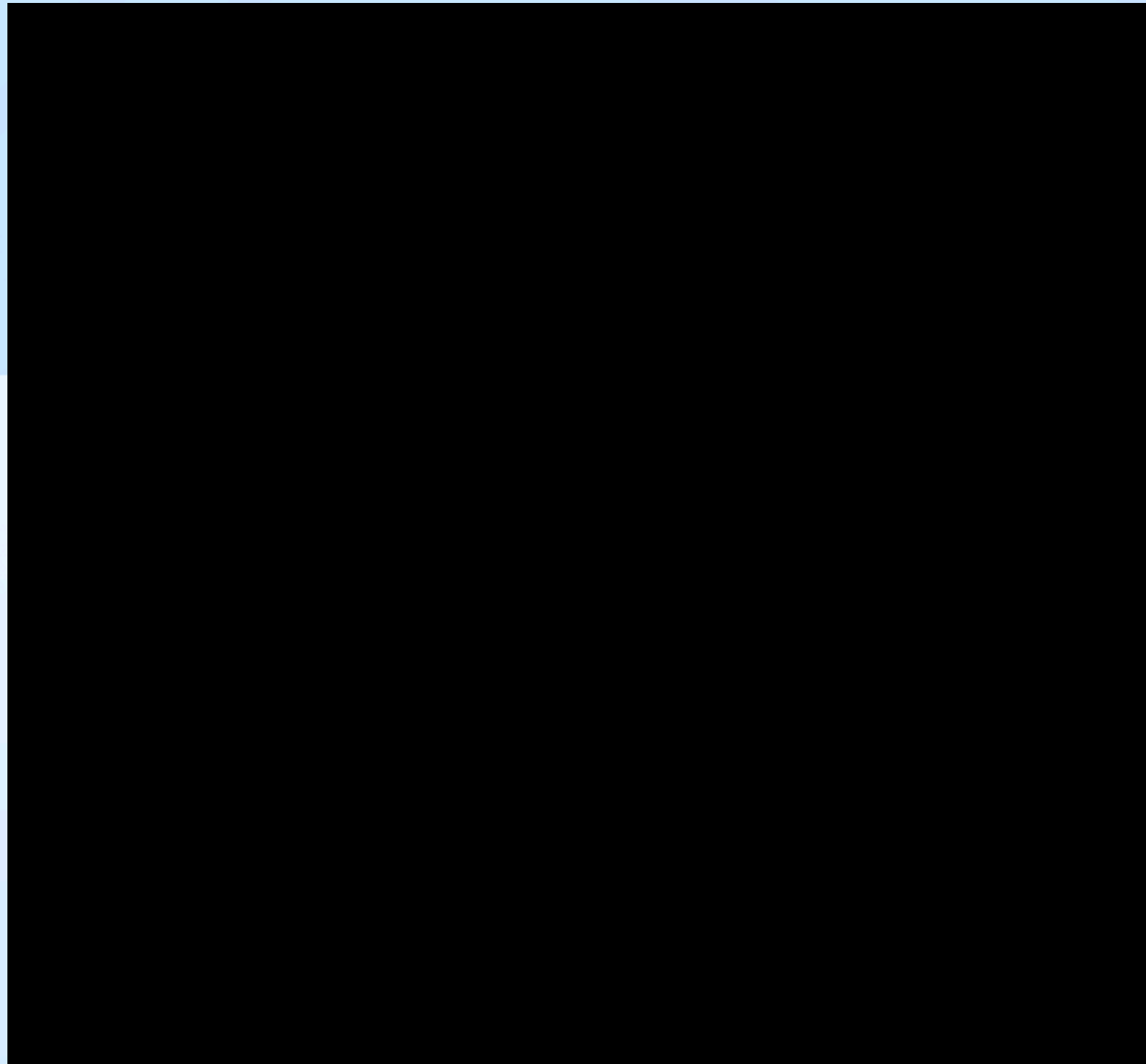
# Evaluation through perineal ultrasound

**Pelvic Floor Maneuver(Arrow for movement)**



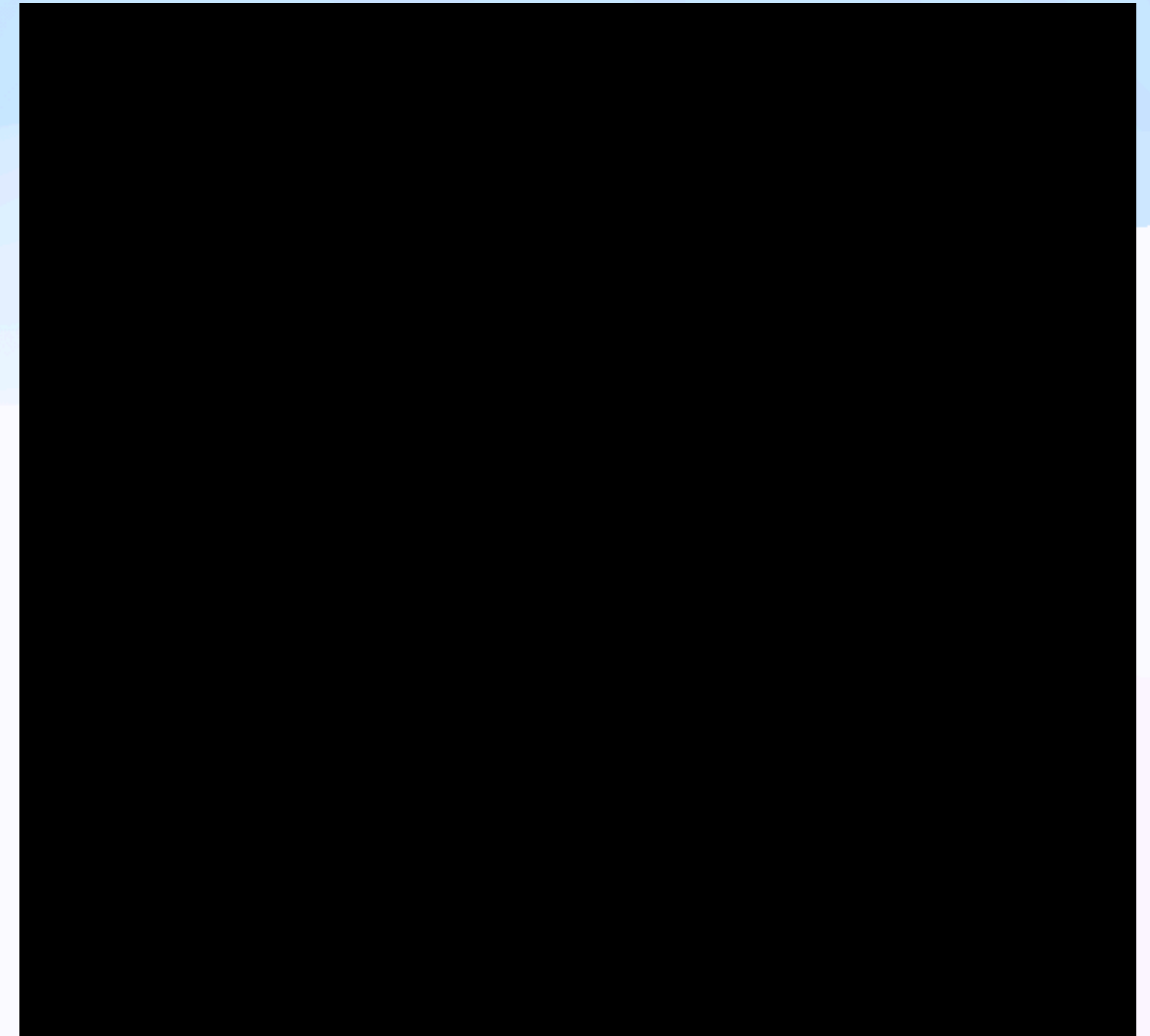
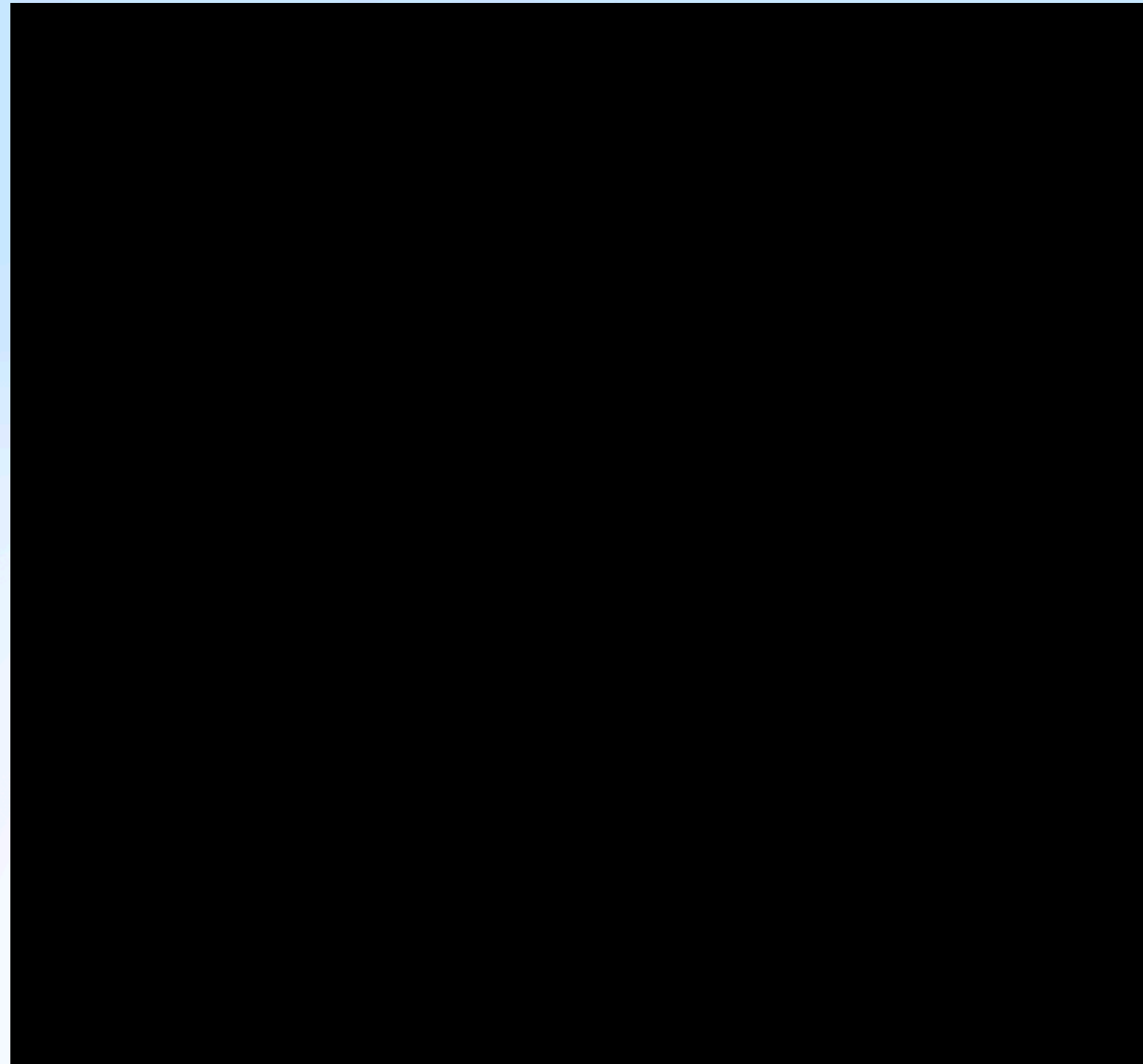
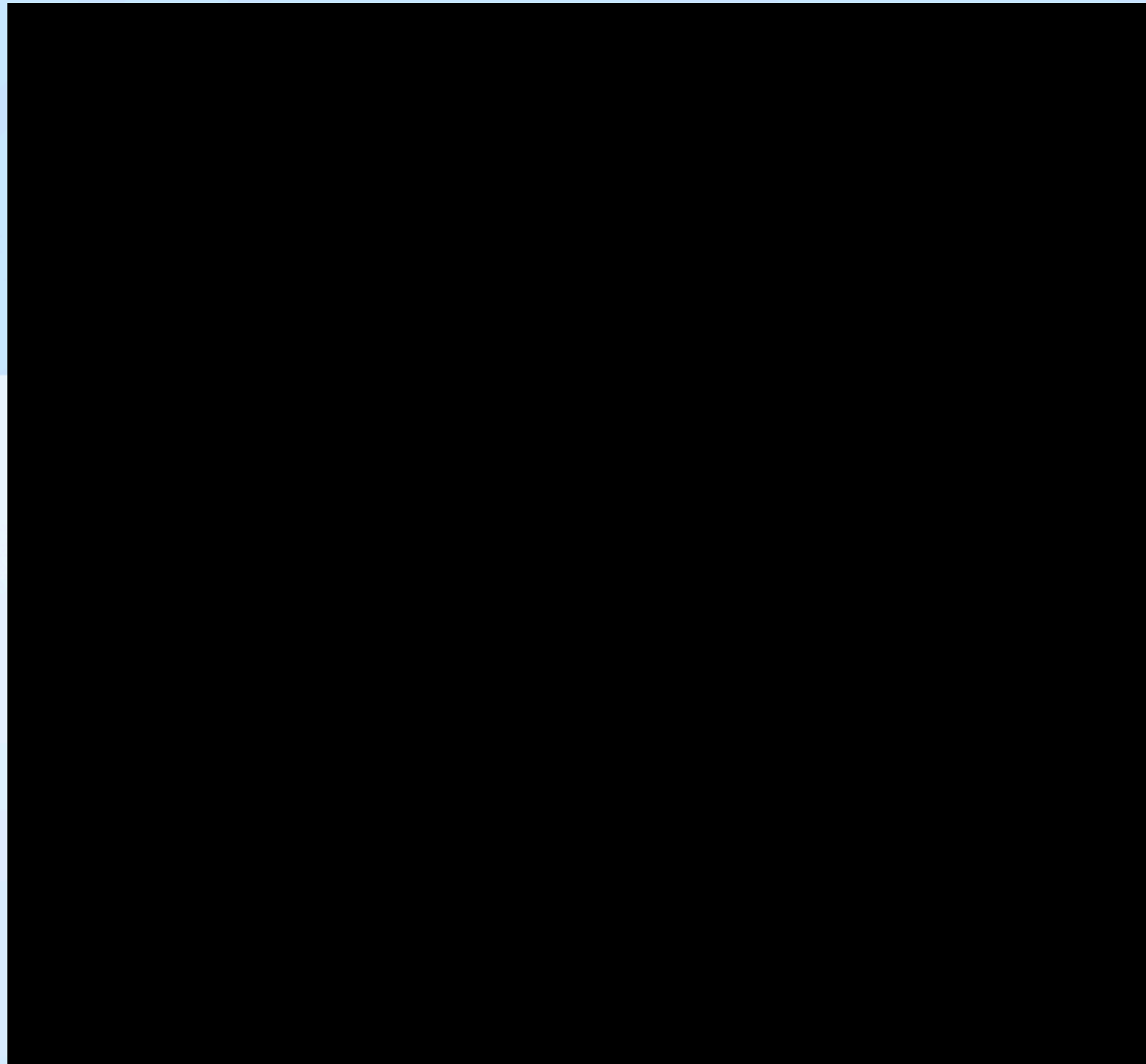
# Evaluation through perineal ultrasound

Valsalva(Arrow for movement)

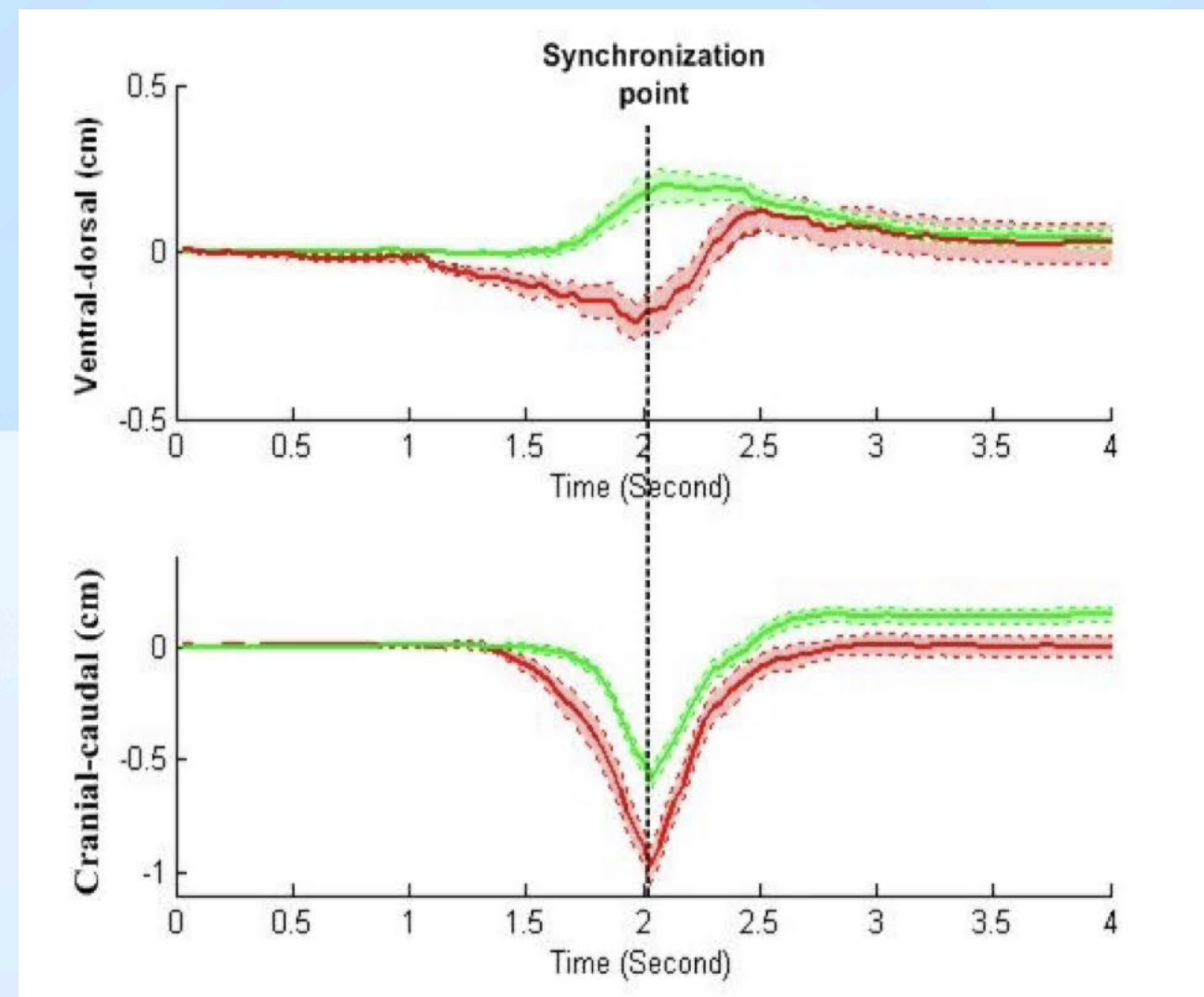


# Evaluation through perineal ultrasound

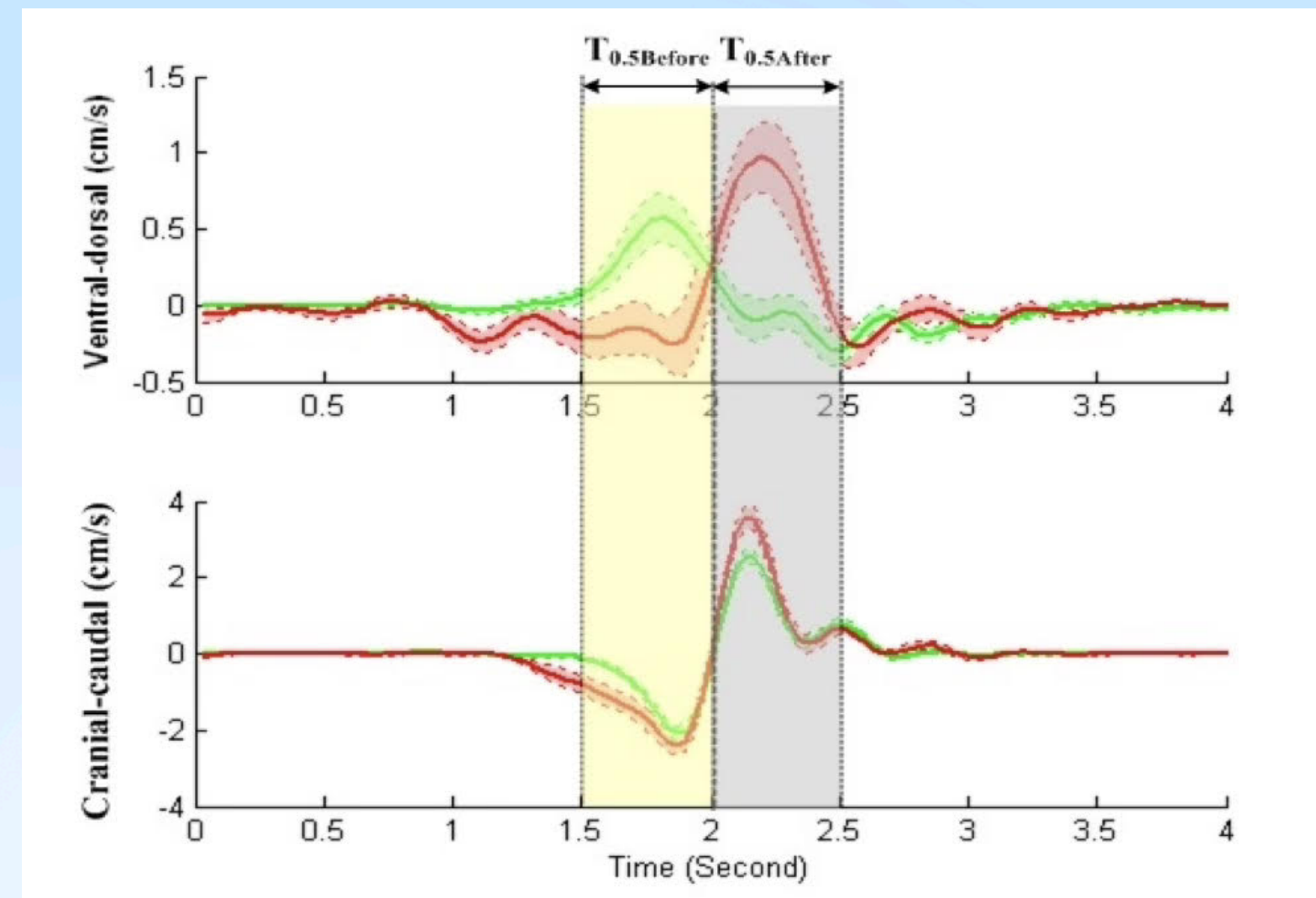
Cough(Arrow for movement)



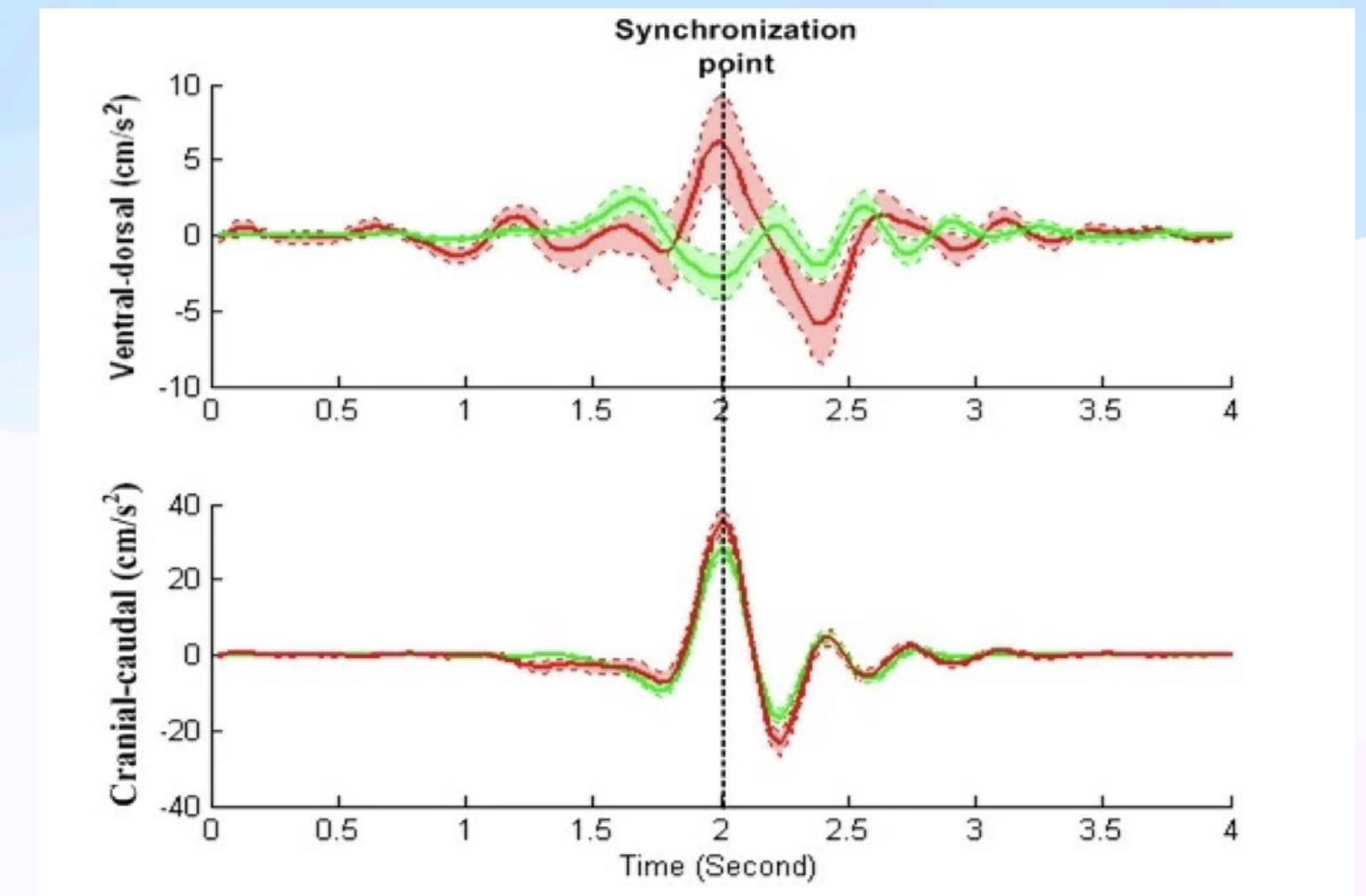
# Evaluation for Cough Induced Stimulation of Physical Quantities



Displacement



Velocity



Acceleration

# Function during Voluntary, Passive, and Transient activation

## Contraction (Active) Valsalva (Passive) and Cough (Transient)

### Voluntary Activation (Contraction)

- Conscious contraction
- Elevates pelvic organs, narrows levator hiatus, increases urethral and anal closure pressure
- Key for strengthening and continence training

### Passive Activation (Valsalva)

- Increased intra-abdominal pressure with downward push
- Tests pelvic floor support and descent
- Used to assess prolapse and organ mobility

### Transient Activation (Cough)

- Rapid, reflexive co-contraction of pelvic floor muscles
- Counters sudden rise in intra-abdominal pressure
- Critical for dynamic continence during daily activities

# Interpretation of Imaging

- Voluntary Pelvic Floor Muscle Contraction results in End Compression of the Urethra
- The results show that this action provides greater support for the urethra
- There are no significant CAUDAL displacement during a cough

# Take Home Message

- Perineal ultrasound scanning provides a non-invasive imaging of dynamic pelvic floor function
- Image analysis documents numerical parameters of passive and dynamic activation of the pelvic floor

# References